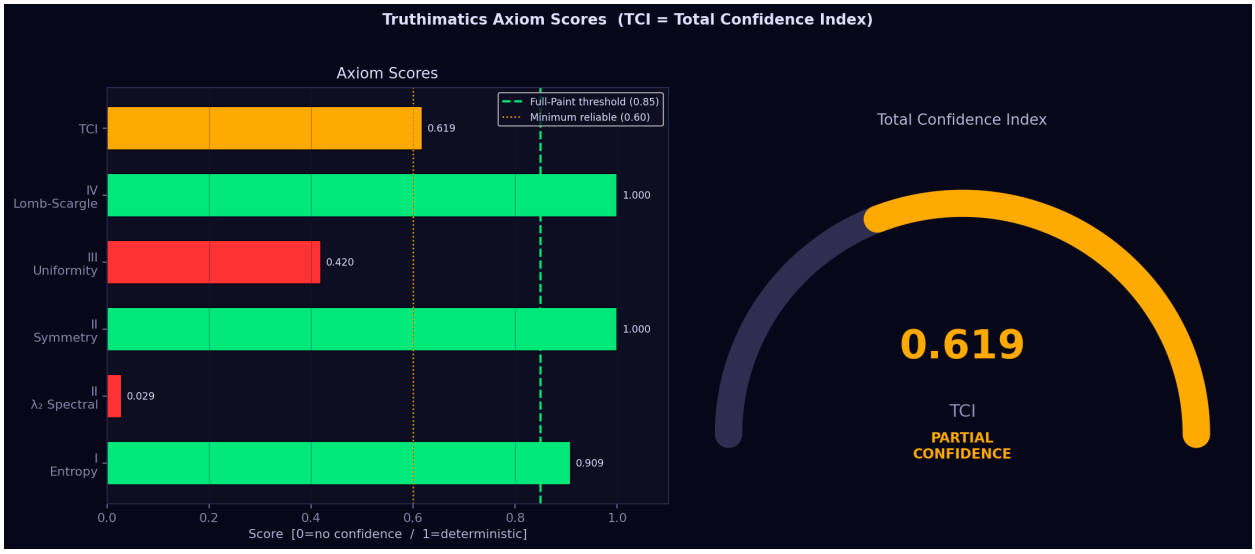
# AXIOM-DINOSAUR v4.0

Truthimatics Full-Paint Deterministic Synthesis

| | |
|---|---|
| **Specimen ID** | AXIOM_AD9803502E |
| **Taxa** | Rhamphorhynchus |
| **Clade** | Pterosaur |
| **PBDB Records** | 30 |
| **Mass (regression)** | 233.1 kg (Campione & Evans 2012) |
| **Mass (volumetric)** | 56.3 kg (Silhouette integral, $\rho$=1060.0 kg/m$^3$) |
| **Body Volume** | 0.072 m$^3$ |
| **Validity Score** | 1.0000 / 1.0000 |
| **TCI** | 0.6185 (partial) |
| **Axiom I Entropy** | 0.9092 |
| **Axiom II $\lambda$■** | 0.1171 (symmetry 100%) |
| **Axiom III Uniformity** | 0.4199 |
| **Axiom IV Lomb FAP** | 1.000000 peak 0.000 Hz |
| **Integrity** | 66.7% Measured / 33.3% Inferred |
| **Elements** | 24 |
| **CoG (m)** | [0.026, 0.177, 0.0] |
| **Generated** | 2026-05-01 00:03 UTC |

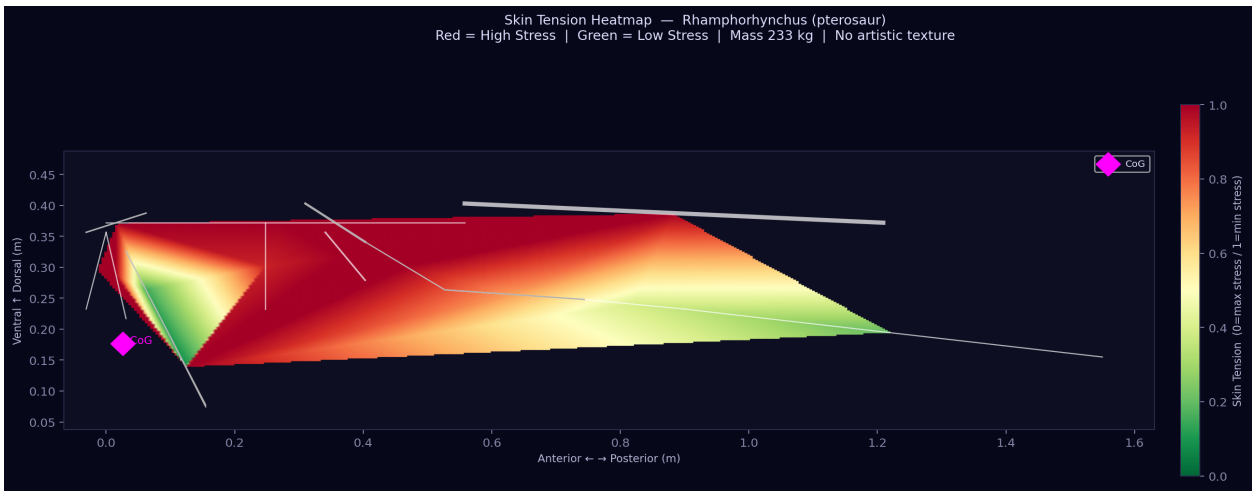
# 1 Truthimatics Confidence Analysis

TCI = weighted sum: Axiom I (entropy, w=0.25) + Axiom II (spectral, w=0.30) + Axiom III (Biewener uniformity, w=0.30) + Axiom IV (Lomb, w=0.15). TCI ≥ 0.85 authorises Full-Paint output.



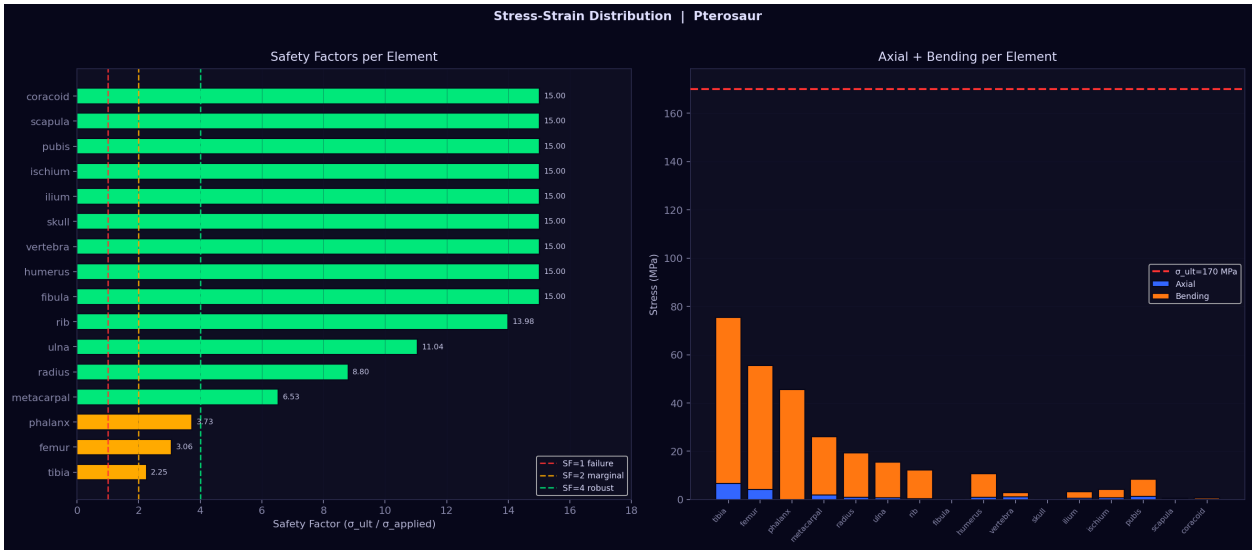
## 2 Skin Tension Heatmap

Stress field interpolated from per-bone safety factors onto the 2D body silhouette. Red=high stress (low SF), Green=low stress (high SF). No artistic texture applied.



# 3 Stress-Strain Distribution

Loading: biped (F=Mg). σ_ult=170 MPa (Reilly & Burstein 1975). SF≥4 robust.

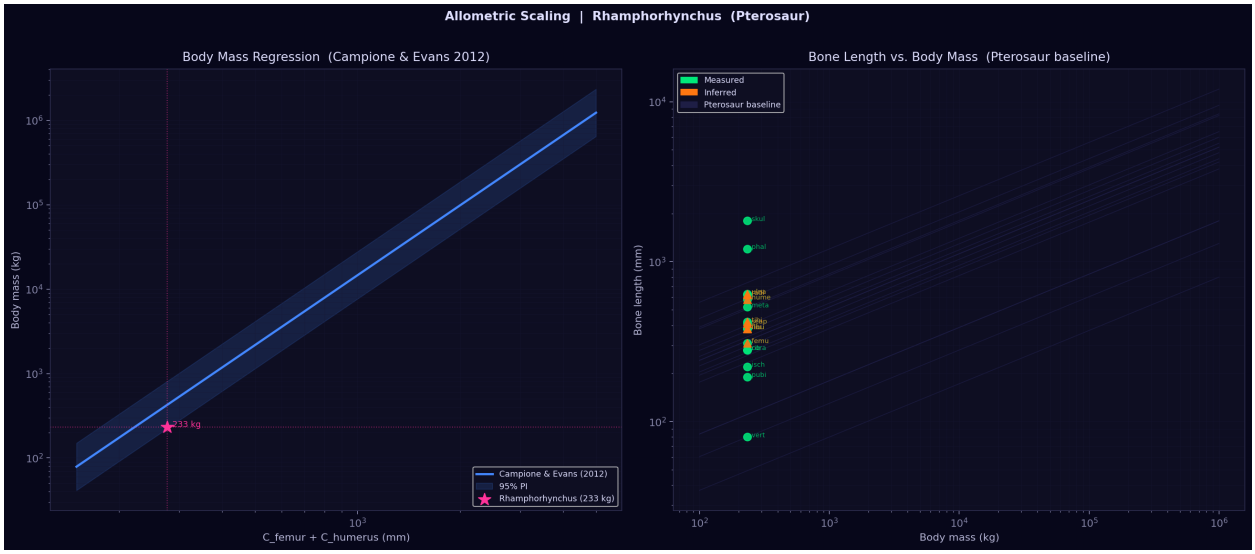



## Per-Element Stress Summary

| Element | σ_ax MPa | σ_bend MPa | σ_total MPa | SF | Status |
|---|---|---|---|---|---|
| coracoid | 0.156 | 0.580 | 0.737 | 15.000 | OK |
| femur | 4.115 | 51.356 | 55.471 | 3.065 | WARNING |
| fibula | 0.000 | 0.000 | 0.000 | 15.000 | OK |
| humerus | 0.896 | 9.765 | 10.661 | 15.000 | OK |
| ilium | 0.574 | 2.621 | 3.194 | 15.000 | OK |
| ischium | 0.706 | 3.500 | 4.206 | 15.000 | OK |
| metacarpal | 1.892 | 24.156 | 26.048 | 6.526 | OK |
| phalanx | 0.000 | 45.627 | 45.627 | 3.726 | WARNING |
| pubis | 1.255 | 7.166 | 8.420 | 15.000 | OK |
| radius | 0.922 | 18.406 | 19.328 | 8.796 | OK |
| rib | 0.373 | 11.785 | 12.159 | 13.982 | OK |
| scapula | 0.093 | 0.392 | 0.485 | 15.000 | OK |
| skull | 0.000 | 0.000 | 0.000 | 15.000 | OK |
| tibia | 6.666 | 68.853 | 75.519 | 2.251 | WARNING |
| ulna | 0.782 | 14.614 | 15.396 | 11.042 | OK |
| vertebra | 1.067 | 1.749 | 2.815 | 15.000 | OK |

# 4 Allometric Scaling Analysis

Campione & Evans (2012): log■■(M_g)=2.754×log■■(C_f+C_h)–1.097 ($R^2$=0.97; N=426 non-avian dinosaurs).



## 5 Physical Properties Analysis

Biomechanical performance estimates derived deterministically from skeletal geometry. Speed: Alexander (1976) locomotion equation. Bite force: Christiansen & Wroe (2007). All values represent physically constrained estimates, not artistic approximations.

| Parameter | Value | Notes |
|---|---|---|
| **Hip Height** | 0.839 m | Femur + tibia + metatarsal |
| **Estimated Speed** | 5.5 km/h  (1.52 m/s) | Gait: Walk \| Fr=0.28 |
| **Stride Length** | 1.314 m | Alexander (1976) / Thulborn (1990) |
| **Bite Force** | 1826.4 N  (186.2 kgf) | Pterosaur regression (Witton 2010) |
| **Skull Length** | 1800.0 mm | Mid-shaft proxy from skull feature vector |
| **Femoral Robusticity** | 0.2838 | Circumference / length (dimensionless) |
| **Limb Strength Index** | 1.3e-05 | (A_f + A_t) / M^0.67  (Christiansen 1998) |
| **Est. Body Length** | 2.79 m | Allometric from femur length |
| **Est. Neck Length** | 0.84 m | Allometric from femur length |
| **Est. Tail Length** | 0.22 m | Allometric from femur length |

## 6 Complete Bone Manifest

| Element | L (mm) | D (mm) | Vol cm³ | Mass g | Method | Conf | SF |
|---|---|---|---|---|---|---|---|
| coracoid | 280.0 | 85.0 | 810.32 | 1011.5 | Measured | 0.88 | 15.00 |
| femur | 310.0 | 28.0 | 97.35 | 121.5 | Measured | 0.88 | 3.06 |
| femur_right | 310.0 | 28.0 | 97.35 | 121.5 | Symmetry | 0.84 | 3.06 |
| fibula | 380.0 | 8.0 | 6.88 | 7.2 | Measured | 0.88 | 15.00 |
| fibula_right | 380.0 | 8.0 | 6.88 | 7.2 | Symmetry | 0.84 | 15.00 |
| humerus | 580.0 | 60.0 | 836.35 | 1044.0 | Measured | 0.88 | 15.00 |
| humerus_right | 580.0 | 60.0 | 836.35 | 1044.0 | Symmetry | 0.84 | 15.00 |
| ilium | 380.0 | 75.0 | 856.18 | 1068.8 | Measured | 0.88 | 15.00 |
| ilium_right | 380.0 | 75.0 | 856.18 | 1068.8 | Symmetry | 0.84 | 15.00 |
| ischium | 220.0 | 40.0 | 140.99 | 176.0 | Measured | 0.88 | 15.00 |
| metacarpal | 520.0 | 45.0 | 157.13 | 129.3 | Measured | 0.88 | 6.53 |
| phalanx | 1200.0 | 30.0 | 130.29 | 100.9 | Measured | 0.88 | 3.73 |
| pubis | 190.0 | 30.0 | 68.49 | 85.5 | Measured | 0.88 | 15.00 |
| radius | 620.0 | 35.0 | 304.22 | 379.8 | Measured | 0.88 | 8.80 |
| radius_right | 620.0 | 35.0 | 304.22 | 379.8 | Symmetry | 0.84 | 8.80 |
| rib | 280.0 | 12.0 | 16.15 | 20.2 | Measured | 0.88 | 13.98 |
| scapula | 410.0 | 110.0 | 1987.14 | 2480.6 | Measured | 0.88 | 15.00 |
| scapula_right | 410.0 | 110.0 | 1987.14 | 2480.6 | Symmetry | 0.84 | 15.00 |
| skull | 1800.0 | 220.0 | 34896.18 | 43560.9 | Measured | 0.88 | 15.00 |
| tibia | 420.0 | 22.0 | 81.42 | 101.6 | Measured | 0.88 | 2.25 |
| tibia_right | 420.0 | 22.0 | 81.42 | 101.6 | Symmetry | 0.84 | 2.25 |
| ulna | 630.0 | 38.0 | 364.39 | 454.9 | Measured | 0.88 | 11.04 |
| ulna_right | 630.0 | 38.0 | 364.39 | 454.9 | Symmetry | 0.84 | 11.04 |
| vertebra | 80.0 | 55.0 | 96.93 | 121.0 | Measured | 0.88 | 15.00 |